

Chapter 7:

Operator Rules, Real Eigenvalues, and Projection Operator

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Operator Rules in the Bra-ket Notation

- We know that an operator is a matrix, e.g. $\mathbf{M}$. An operator in quantum computing rotates the vector (a quantum state), $|\alpha\rangle$, in a hyperspace. So, when it operates on a vector, it gives another vector (i.e. a rotated vector), $|\alpha'\rangle$.

$$|\alpha'\rangle = \mathbf{M}(|\alpha\rangle) = \mathbf{M}|\alpha\rangle$$

- If there are two operators, $\mathbf{M}$ and $\mathbf{N}$, that give the same final vector after rotating the same vector, $|\alpha\rangle$, for all $|\alpha\rangle$ in the space, then they must be the same,

$$\mathbf{M}|\alpha\rangle = \mathbf{N}|\alpha\rangle, \forall |\alpha\rangle \Rightarrow \mathbf{M} = \mathbf{N}$$

This is true unless the space only contains $|0\rangle$. $|0\rangle$ is the so-called **null vector**. It is just like the zero in real number that anything multiplied by it will be a $|0\rangle$.

- For example, in a 2-D space, $|0\rangle = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. In a 4-D space, $|0\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$.
- Their corresponding bra vectors are $\langle 0| = (0 \ 0)$ and $\langle 0| = (0 \ 0 \ 0 \ 0)$ in the 2-D spaces and 4-D spaces.
- As a matrix, the operator should obey distribution law also, $\mathbf{M}(\mathbf{C}_\alpha|\alpha\rangle + \mathbf{C}_\beta|\beta\rangle) = \mathbf{C}_\alpha\mathbf{M}|\alpha\rangle + \mathbf{C}_\beta\mathbf{M}|\beta\rangle$. where $\mathbf{C}_\alpha$ and $\mathbf{C}_\beta$ are scalars (any complex numbers).

Operator Rules in the Bra-ket Notation

- Due to dual correspondence, a vector in the ket space, $|\alpha\rangle$, has a corresponding vector in the bra space, $\langle\alpha|$. Then the rotated vector in the ket space, $|\alpha'\rangle$, must also have a corresponding vector in the bra space $\langle\alpha'|$:

$$\langle\alpha|M^\dagger = \langle\alpha'| \Leftrightarrow M|\alpha\rangle = |\alpha'\rangle$$

- We can also say M is the adjoint of $M^\dagger$ because $M = (M^\dagger)^\dagger$. So, the operators in the ket and bra spaces are the adjoint of each other.
- Another important rule is the **associative axiom**. It can be written as below equation, the left-hand side is the inner product of two vectors, $|\alpha\rangle$ and $M|\beta\rangle$, and the product is a scalar. $M|\beta\rangle$ is a vector because it is the rotation of $|\beta\rangle$ in the hyperspace by the operator M .

$$(\langle\alpha|)(M|\beta\rangle) = ((\langle\alpha|M)|\beta\rangle)$$

- For example, $|\beta\rangle = \begin{pmatrix} i \\ 1 \end{pmatrix}$, $M = \begin{pmatrix} i & i \\ 0 & 1 \end{pmatrix}$, $|\alpha\rangle = \begin{pmatrix} 2 \\ i \end{pmatrix}$. Then, $M|\beta\rangle = \begin{pmatrix} i & i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} i \\ 1 \end{pmatrix} = \begin{pmatrix} -1+i \\ 1 \end{pmatrix}$ and, thus, $(\langle\alpha|)(M|\beta\rangle) = (2 \ -i) \begin{pmatrix} -1+i \\ 1 \end{pmatrix} = -2 + i$, which is a scalar complex number.
- The right-hand side is also the inner product of two vectors. It is the multiplication of the bra space vector $\langle\alpha|M$ and the ket space vector $|\beta\rangle$, and the product is a scalar.

Operator Rules in the Bra-ket Notation

- The corresponding bra space vector must be $\langle \mathbf{M}\beta |$ by definition. Therefore, we will get $\langle \alpha | \mathbf{M}\beta \rangle = \langle \mathbf{M}\beta | \alpha \rangle^*$.
- We also know that due to dual correspondence, the corresponding bra vector of $\mathbf{M}|\beta\rangle$ is also $\langle \beta | \mathbf{M}^\dagger$.
So, we can arrange the above equation as:

$$\langle \alpha | \mathbf{M} | \beta \rangle = \langle \beta | \mathbf{M}^\dagger | \alpha \rangle^*$$

- The notation of $\langle \alpha | \mathbf{M} | \beta \rangle$ is just the multiplication of a row vector to a matrix and then to a column vector from the left to the right in the form of:

$$(\blacksquare \cdots \blacksquare) \begin{pmatrix} \blacksquare & \cdots & \blacksquare \\ \vdots & \ddots & \vdots \\ \blacksquare & \cdots & \blacksquare \end{pmatrix} \begin{pmatrix} \blacksquare \\ \vdots \\ \blacksquare \end{pmatrix}$$

Eigenvalues of Hermitian Matrix

- Hermitian matrices are very important in quantum computing because Hermitian matrices have real eigenvalues.
- Matrices correspond to operators and some of them even correspond to measurements (e.g. Pauli spin matrices) and observables.
- The measured values are the values we read in our daily life and must be real numbers (e.g. ± 1). Therefore, all operators corresponding to observables and measurements must be Hermitian.
- If a matrix $\mathbf{M}$ is Hermitian, it means $\mathbf{M}^\dagger = \mathbf{M}$. If λ_i and $|i\rangle$ are its i -th eigenvalue and eigenvector, respectively, we have

$$\mathbf{M}|i\rangle = \lambda_i|i\rangle$$

- By finding the inner product of vector $\mathbf{M}|i\rangle$ (This is the rotation of $|i\rangle$ by $\mathbf{M}$ and it is a vector) and $\langle i|$.

$$\langle i|\mathbf{M}|i\rangle = \langle i|\lambda_i|i\rangle = \lambda_i\langle i|i\rangle$$

- We can pull λ_i out because it is just a scalar and $\langle i|i\rangle$ is 1 if we assume $\mathbf{M}$'s eigenvector are normalized.

$$\langle i|\mathbf{M}|i\rangle = \langle i|\mathbf{M}^\dagger|i\rangle^*$$

- Since $\mathbf{M}$ is Hermitian,

$$\langle i|\mathbf{M}|i\rangle = \langle i|\mathbf{M}|i\rangle^*$$

We thus have $\lambda_i = \lambda_i^*$. And λ_i must be real as it equals to its complex conjugate.

Copenhagen Interpretation/Born Rule and Projection Operator

- The operator that corresponds to an observable is required to be Hermitian to be meaningful so that it will have real eigenvalues. This is related to the **Copenhagen Interpretation**.
- It states that “*If an observable corresponding to a self-adjoint operator (i.e. Hermitian) $\mathbf{M}$ with discrete spectrum is measured in a system with normalized state Ψ , then*
 1. *The result will be one of the eigenvalues λ_i of $\mathbf{M}$.*
 2. *The probability is $\langle \Psi | P_i | \Psi \rangle$, where P_i is projection operator to project any state to the subspace of $|i\rangle$.*”
- The first statement tells us that when we perform a measurement corresponding to $\mathbf{M}$, it must collapse to an eigenstate, $|i\rangle$, corresponding to one of the eigenvalues, λ_i , of $\mathbf{M}$ (e.g. either spin up or down in the σ_z case).
- The second statement tells us the probability of a certain state we will get. We learned that the probability $|\psi\rangle$ is expressed as the linear combination of the basis states formed by the eigenstates of $\mathbf{M}$.
- For example, $|\psi\rangle = a_0|0\rangle + a_1|1\rangle + \dots + a_{N-1}|N-1\rangle$, the probability to get state $|i\rangle$ is just $|a_i|^2$.
- In the above statement, it gives a more general equation to find the probability, i.e. $\langle \psi | P_i | \psi \rangle$, by introducing the **projection operator, P_i** .

Copenhagen Interpretation/Born Rule and Projection Operator

- The projection operator is just a matrix. What it does is to project a state into a subspace. In our real 3D space, a building has a 3D nature that has height, width, and depth. We can project its shadow on the floor, maybe the 2D x-y plane, or on a vertical wall, maybe the 2D y-z plane.
- $|\psi\rangle$ is a vector in a high dimensional space. We can project it to a lower dimension space using an appropriate projection operator. Here P_i is the projection operator that will project an arbitrary vector $|\psi\rangle$ to the subspace spanned by $|i\rangle$.
- The projection operator that projects a vector to a subspace formed by $|i\rangle$ is given by $P_i = |i\rangle\langle i|$.
- A ket multiplied by a bra is a matrix and this is called the **outer product** or **tensor product** of vectors.
- We write it in matrix form but then we need to choose a set of basis states first. The easiest one is to choose the eigenstates of M as the basis states.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \quad \dots \quad |N-1\rangle = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Copenhagen Interpretation/Born Rule and Projection Operator

- So $|i\rangle$ has the i -th entry non-zero in a column vector and $\langle i|$ has the i -th entry non-zero in a row vector. Let's assume $N = 4$ and $i = 2$, we have

$$P_i = P_2 = |2\rangle\langle 2|$$

$$= \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} (0 \ 0 \ 1 \ 0) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- Let us see how P_i accomplish the projection task by apply P_i to $|\psi\rangle$:

$$P_i|\psi\rangle = P_i(a_0|0\rangle + a_1|1\rangle + \cdots + a_{N-1}|N-1\rangle)$$

- We can distribute P_i into each term as follows:

$$\begin{aligned} P_i|\psi\rangle &= P_i(a_0|0\rangle + a_1|1\rangle + \cdots + a_{N-1}|N-1\rangle) \\ &= a_0P_i|0\rangle + a_1P_i|1\rangle + \cdots + a_{N-1}P_i|N-1\rangle \\ &= a_0|i\rangle\langle i||0\rangle + a_1|i\rangle\langle i||1\rangle + \cdots + a_{N-1}|i\rangle\langle i||N-1\rangle \end{aligned}$$

Copenhagen Interpretation/Born Rule and Projection Operator

- Due to the association rule, we get $a_j|i\rangle\langle i|j\rangle$ for $j = 0$ to $N - 1$. Since the basis states are supposed to be orthonormal, only $\langle i|i\rangle$ is non-zero. Therefore, $P_i|\psi\rangle = a_i|i\rangle\langle i|i\rangle = a_i|i\rangle$.
- The projection operation removes all the orthogonal components and only keeps the component in the subspace spanned by $|i\rangle$.
- Then we apply the bra vector $\langle\psi|$ from the left,

$$\begin{aligned}
 \langle\psi|P_i|\psi\rangle &= \langle\psi|(P_i|\psi\rangle) = \langle\psi|(a_i|i\rangle) \\
 &= (a_0^*\langle 0| + a_1^*\langle 1| + \dots + a_{N-1}^*\langle N-1|)a_i|i\rangle \\
 &= a_0^*a_i\langle 0|i\rangle + a_1^*a_i\langle 1|i\rangle + \dots + a_{N-1}^*a_i\langle N-1|i\rangle \\
 &= a_i^*a_i = |a_i|^2
 \end{aligned}$$

- What if we apply P_i twice?

$$\begin{aligned}
 P_iP_i|\psi\rangle &= P_i|i\rangle\langle i||\psi\rangle = |i\rangle\langle i||i\rangle\langle i||\psi\rangle \\
 &= |i\rangle(\langle i|i\rangle)\langle i||\psi\rangle = |i\rangle(1)\langle i||\psi\rangle = P_i|\psi\rangle
 \end{aligned}$$

Summary

- **Operator Application:** We have learned how to apply operators within the bra-ket (Dirac) notation.
- **Associative Axiom:** You understand the utility of the associative property in mathematics to simplify complex expressions by performing easier operations first.
- **Hermitian Matrices:** You have seen the proof that Hermitian matrices always have real eigenvalues, a fundamental property in quantum mechanics.
- **Projection Operator:** The concept of the projection operator has been introduced.
- **Outer Product:** The outer product concept has been introduced, which, along with the projection operator, will be crucial for understanding future topics in quantum computing.

Chapter 8:

Eigenvalue, Matrix Diagonalization and

Unitary Matrix

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Eigenvalues and Matrix Diagonalization

- The concept of matrix diagonalization is related to the change of basis.
- For any matrix, A , with dimension $n \times n$, it can be written as the following with n rows and n columns.

$$A = \begin{pmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,n-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,n-1} \end{pmatrix} = (a_{i,j})$$

- We index the matrix elements using the row-column convention. We may also write it as $(a_{i,j})$ to indicate that the matrix has the i -th row and j -th column element equals to $a_{i,j}$.
- To find its eigenvalues λ_i , and eigenvectors, $|i\rangle$, we need to solve the following equation, $A|i\rangle = \lambda_i|i\rangle$. The eigenvector of a matrix is so special because when we apply the matrix to that vector, the vector is not rotated but just scaled by a scalar.

Eigenvalues and Matrix Diagonalization

- For example, $|i\rangle$ may be $\begin{pmatrix} 1 \\ 2 \\ 0 \\ 5 \end{pmatrix}$ if $n = 4$. In general, $|i\rangle = \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{n-1} \end{pmatrix}$ for an n -dimensional vector so, we can write:

$$A|i\rangle = \begin{pmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,n-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,n-1} \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{n-1} \end{pmatrix} = \lambda_i \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{n-1} \end{pmatrix}$$

- The right-hand side is the multiplication of a scalar and a vector. But we can also write it as a matrix-vector multiplication,

$$\begin{aligned} A|i\rangle &= \begin{pmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,n-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,n-1} \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{n-1} \end{pmatrix} \\ &= \begin{pmatrix} \lambda_i & 0 & \cdots & 0 \\ 0 & \lambda_i & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_i \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{n-1} \end{pmatrix} = \lambda_i \mathbf{I} |i\rangle \end{aligned}$$

- We see that a scalar multiplied to a vector acts the same as a diagonal matrix with the scalar value as the diagonal element multiplied to that vector. This is because $\lambda_i |i\rangle = \lambda_i \mathbf{I} |i\rangle$, where $\mathbf{I}$ is the identity matrix.

Eigenvalues and Matrix Diagonalization

- If we move the right-hand side term to the left, we have

$$(\mathbf{A} - \lambda_i \mathbf{I})|\mathbf{i}\rangle = \begin{pmatrix} a_{0,0} - \lambda_i & a_{0,1} & \cdots & a_{0,n-1} \\ a_{1,0} & a_{1,1} - \lambda_i & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,n-1} - \lambda_i \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{n-1} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

- The right-hand side is the null vector $\mathbf{0}$. We need to equate the left to the right and there are n equations. For the j -th row, we need to equate

$$(a_{j,0})\alpha_0 + (a_{j,1})\alpha_1 + \cdots + (a_{j,j} - \lambda_i)\alpha_j + \cdots + (a_{j,n-1})\alpha_{n-1} = 0$$

- If all α_i 's equal to zero, i.e. $|\mathbf{i}\rangle = \mathbf{0}$, that is a valid solution but this is useless to us. We want to find non-zero $|\mathbf{i}\rangle$ which will be the eigenvectors of $\mathbf{A}$. We have n equations and n variables (α_0 to α_{n-1}). The theory tells us that we need to make the determinant of the matrix zero.

$$|\mathbf{A} - \lambda_i \mathbf{I}| = \begin{vmatrix} a_{0,0} - \lambda_i & a_{0,1} & \cdots & a_{0,n-1} \\ a_{1,0} & a_{1,1} - \lambda_i & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,n-1} - \lambda_i \end{vmatrix} = \mathbf{0}$$

Eigenvalues and Matrix Diagonalization

- This is equivalent to what we did in Chap. 5. We want to find the eigenvalues and eigenvectors of σ_x .
- Since $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, the equation we need to solve is

$$\begin{vmatrix} a_{0,0} - \lambda_i & a_{0,1} \\ a_{1,0} & a_{1,1} - \lambda_i \end{vmatrix} = \begin{vmatrix} 0 - \lambda_i & 1 \\ 1 & 0 - \lambda_i \end{vmatrix} = 0$$

- The determinant is thus $-\lambda_i \times -\lambda_i - 1 \times 1$. Therefore, the equation becomes, $\lambda_i^2 - 1 = 0$. And thus, $\lambda_i = \pm 1$. We can assign, $\lambda_0 = 1$ and $\lambda_1 = -1$. This is what we got in Chap. 5. Now we know the eigenvalues, we can use them to find the eigenvectors. For $\lambda_0 = 1$, we have

$$\begin{pmatrix} a_{0,0} - \lambda_i & a_{0,1} \\ a_{1,0} & a_{1,1} - \lambda_i \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} = \begin{pmatrix} 0 - 1 & 1 \\ 1 & 0 - 1 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This give us two equations,

$$\begin{aligned} (0 - 1)\alpha_0 + (1)\alpha_1 &= 0 \\ (1)\alpha_0 + (0 - 1)\alpha_1 &= 0 \end{aligned}$$

- So, for $\lambda_0 = 1$, the eigenvector is $|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. And for $\lambda_0 = -1$, the eigenvector is $|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Eigenvalues and Matrix Diagonalization

- In general, if we represent a matrix using its eigenvectors as the basis vectors, it will be a diagonal matrix with the eigenvalues as the diagonal elements. This process is called diagonalization.

$$A = \begin{pmatrix} \lambda_0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_{n-1} \end{pmatrix}$$

- But how do we do that? We need to find its eigenvectors and eigenvalues first. Therefore, the diagonalization of a matrix is equivalent to finding its eigenvectors and eigenvalues.
- *It is also important to note that while the matrix and vector representations change with basis, the eigenvalues are kept the same.*

Unitary Matrix

- In quantum computing and quantum mechanics, the unitary matrix is very important. This is as important as the Hermitian matrix.
- A matrix is an operator that rotates the vector (state). A Hermitian matrix is a matrix that is equal to its adjoint matrix, i.e. $A^\dagger = A$, where $A^\dagger$ is the adjoint matrix of A by performing the complex conjugate and transpose operations.
- The **Hermitian matrix** is important because it has real eigenvalues and therefore, all observables must correspond to Hermitian matrices.
- The **unitary matrix** is important because it preserves the inner product of vectors when they are transformed together by a unitary matrix. It also preserves the length of a vector. This is very important because it will preserve the probability amplitude of a vector in quantum computing so that it is always 1.
- A unitary matrix, U , is defined as a matrix whose adjoint, $U^\dagger$, equals to its inverse, U^{-1} (instead of to itself as in the Hermitian matrix).

$$U^\dagger = U^{-1}$$

If we multiply both sides by U from the right or the left, respectively, we get

$$U^\dagger U = U U^\dagger = I$$

where I is the identity matrix. This is because any matrix multiplied by its inverse must equal the identity matrix by definition.

Unitary Matrix

- Why this is important? Assume there are two vectors, $|f\rangle$ and $|g\rangle$. Their inner product is $\langle f|g\rangle$.
- What happens if we operate U on the vectors (i.e. rotate them in hyperspace)? We expect their inner product to be the same. For example, your table's corner angle should be the same if you rotate your table (recall that the inner product is related to the angle between the two edges in the real space).
- If two basis vectors are orthogonal to each other (meaning they will not appear at the same time during a measurement), they are still orthogonal after being rotated together.
- After applying U , $|f\rangle$ and $|g\rangle$ become $U|f\rangle$ and $U|g\rangle$, respectively. Their inner product is the multiplication of the bra version of $U|f\rangle$ on the ket version of $U|g\rangle$. So, we get

$$\langle f|U^\dagger U|g\rangle = \langle f|I|g\rangle = \langle f|g\rangle$$

- We see that the inner product is conserved if the rotation matrix is unitary.
- In quantum computing, when we operate a qubit, we need to make sure the operation matrix is unitary.
- The unitary matrix is also used to describe the evolution of a quantum state as a function of time.

Unitary Matrix

- Assume

$$\mathbf{U} = \begin{pmatrix} b_{0,0} & b_{0,1} & \cdots & b_{0,n-1} \\ b_{1,0} & b_{1,1} & \cdots & b_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n-1,0} & b_{n-1,1} & \cdots & b_{n-1,n-1} \end{pmatrix}$$

where it has n columns and n rows. So, $\mathbf{U}^\dagger \mathbf{U}$ becomes:

$$\begin{aligned} \mathbf{U}^\dagger \mathbf{U} &= \begin{pmatrix} b_{0,0}^* & b_{1,0}^* & \cdots & b_{n-1,0}^* \\ b_{0,1}^* & b_{1,1}^* & \cdots & b_{n-1,1}^* \\ \vdots & \vdots & \ddots & \vdots \\ b_{0,n-1}^* & b_{1,n-1}^* & \cdots & b_{n-1,n-1}^* \end{pmatrix} \begin{pmatrix} b_{0,0} & b_{0,1} & \cdots & b_{0,n-1} \\ b_{1,0} & b_{1,1} & \cdots & b_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n-1,0} & b_{n-1,1} & \cdots & b_{n-1,n-1} \end{pmatrix} \\ &= \mathbf{I} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \end{aligned}$$

In general, $I_{i,j} = b_{0,i}^* b_{0,j} + b_{1,i}^* b_{1,j} + \cdots + b_{n-1,i}^* b_{n-1,j} = \delta_{i,j}$

- Since only the diagonal element in $\mathbf{I}$ is non-zero and equal to one, therefore, $\delta_{i,j}$ is used.

Unitary Matrix

- If we treat every column of U as a vector and called the i -th column $|v_i\rangle$, we have

$$|v_i\rangle = \begin{pmatrix} b_{0,i} \\ b_{1,i} \\ \vdots \\ b_{n-1,i} \end{pmatrix} \text{ and } \langle v_i| = (b_{0,i}^*, b_{1,i}^*, \dots, b_{n-1,i}^*)$$

Here we take the transpose and conjugate to get the bra version of $|v_i\rangle$. It is just the inner product of the column vectors of U ,

$$I_{i,j} = b_{0,i}^* b_{0,j} + b_{1,i}^* b_{1,j} + \dots + b_{n-1,i}^* b_{n-1,j} = \langle v_i|v_j\rangle$$

However, for the identity matrix, only the diagonal elements are one and the others are zero. That means

$$\langle v_i|v_j\rangle = \delta_{i,j}$$

This Kronecker delta ($\delta_{i,j}$) tells us that the inner products are zero if they are not from the same column and one if they are from the same column (i.e. same vector). Therefore, the column vectors of any unitary matrix are orthonormal to each other. And this is the same for the row vectors in the unitary matrix.

Summary

- **Finding Eigenvalues:** A new, often more convenient, method for finding the eigenvalues of a matrix was derived.
- **Diagonalization:** If a matrix is represented using its own eigenvectors as the basis, it becomes a diagonal matrix where the diagonal elements are its eigenvalues.
- **Unitary Matrix:** The concept of a unitary matrix was introduced, which is crucial in quantum computing.
- **Properties of Unitary Matrices:**
 - They preserve the inner product between any two vectors after transformation.
 - They preserve the magnitude (norm) of a vector.
 - Their rows and columns form an orthonormal set.
- **Unitary Test:** A key method to determine if a matrix U is unitary is to check if it transforms an orthonormal set of basis vectors into another orthonormal set. This principle will be used later to verify quantum oracles.